



Cool and Hot Cars: the Importance of Vehicles Albedo in Urban Environment

António Lopes¹, Ana Rita Seixas^{1,2}, Ana Teves^{1,2}, Ana Inês Simões^{1,2} and Cátia Avelar^{1,2}

¹Institute of Geography and Spatial Planning (IGOT-ULisboa), Center of Geographical Studies, Universidade de Lisboa, Portugal

²Master Students in Physical Geography and GIS

e-mail: antonio.lopes@campus.ul.pt

I – Starting questions:

Q1:How many vehicles may contain a city?

Color and size matters?

Q2:What is the importance of car surfaces in the urban climate?

Q3:How can we include mobile urban structure in the heat fluxes?

Q4:Are they sufficient important to modify the Urban Heat Island pattern?

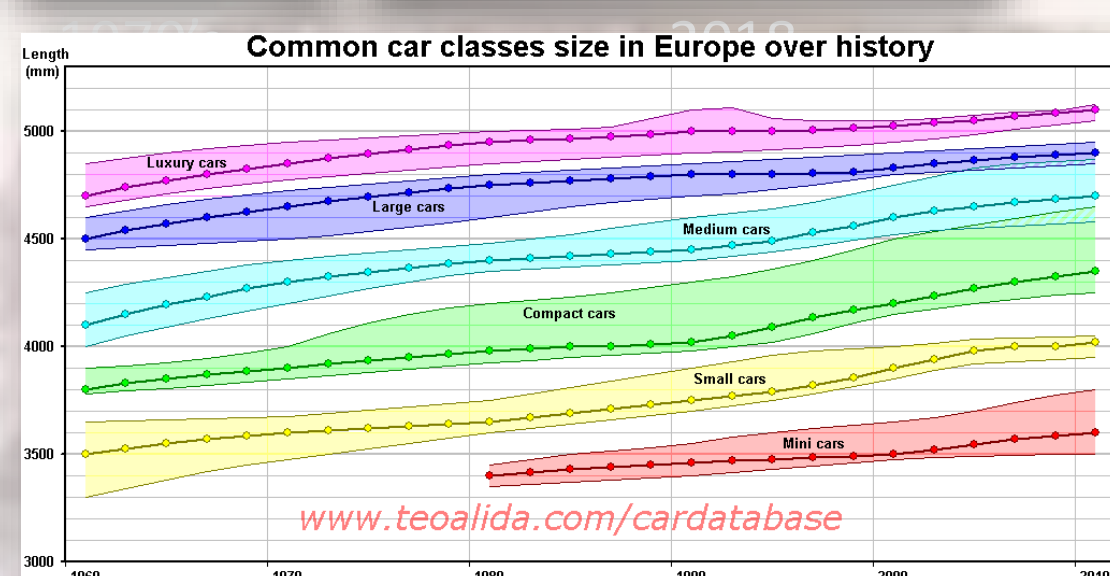
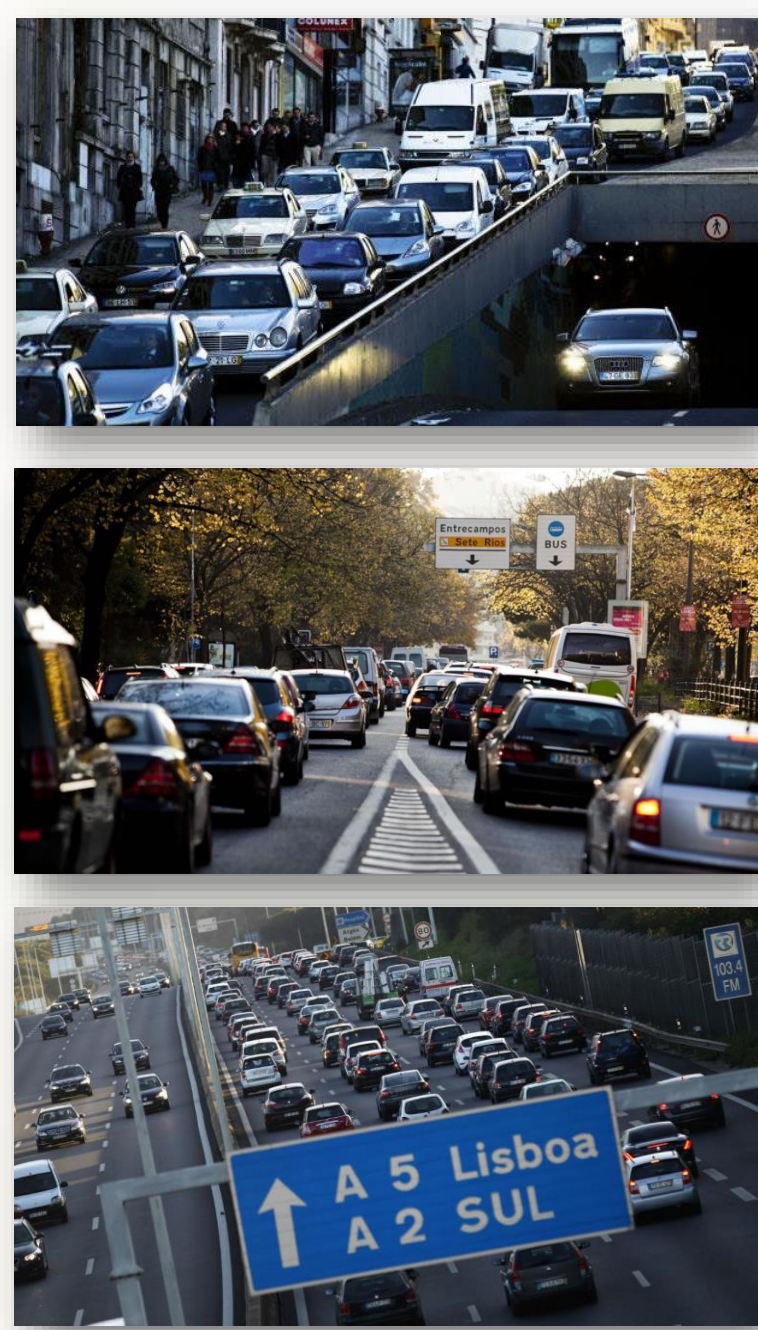


Facts:

Land cover and urban features are variables of upmost importance for urban climate assessment. Researches about UHI commonly associate thermal patterns and urban forms, such as H/W ratio, aerodynamic roughness (z_0) and surface characteristics like albedo, emissivity, surface temperature, etc., that depend on materials and colors among many other factors. In recent years, the number of vehicles has increased in many cities of Europe, with the new economic cycle.

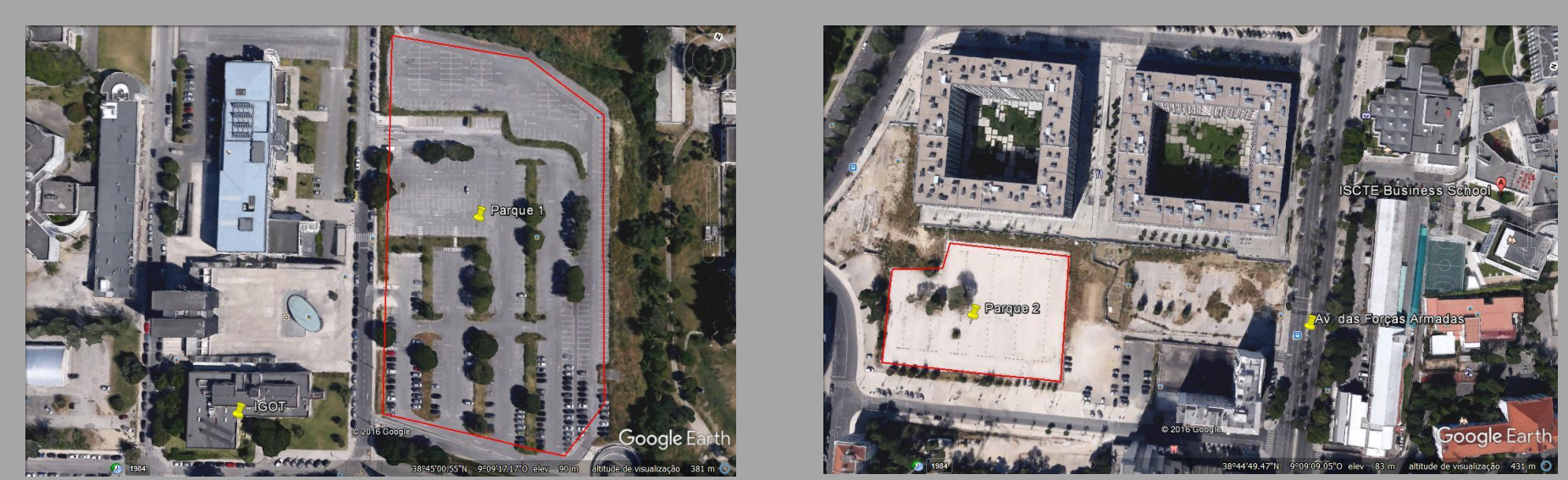
New vehicles design are imposed by comfort and energy efficiency but some commercial strategies lead to an increase in their length: about 2 cm/year in each class bring to cities larger an massive vehicles.

www.teoalida.com/cardatabase/car-length-evolution/



II b – Methods and Data

Two parking lots were chosen (Asphalt and Macadam)



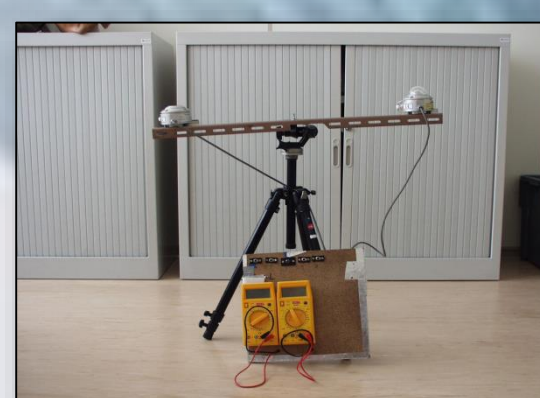
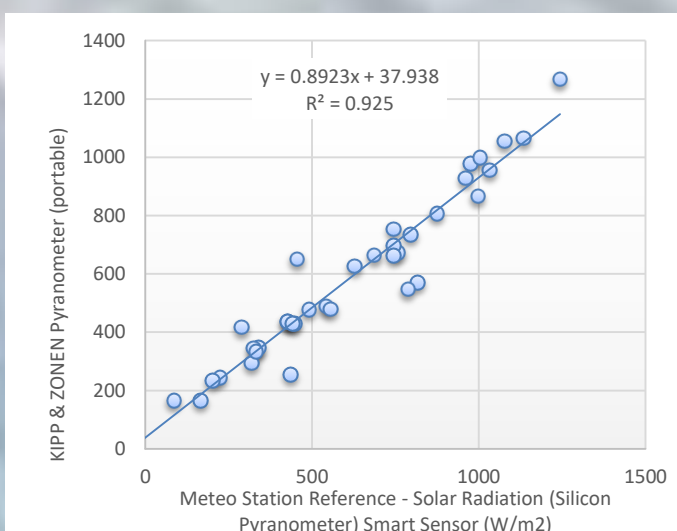
16 Nov measuring time: 12:00h-16:00h

Park 1
Air Temp: 21.7°C
RH: 28.6%
u max: 2.9m/s
u: 0.7 m/s

Park 2
Air Temp: 27.5°C
RH: 19.5%
u max: 1.1 m/s
u: 0.2 m/s

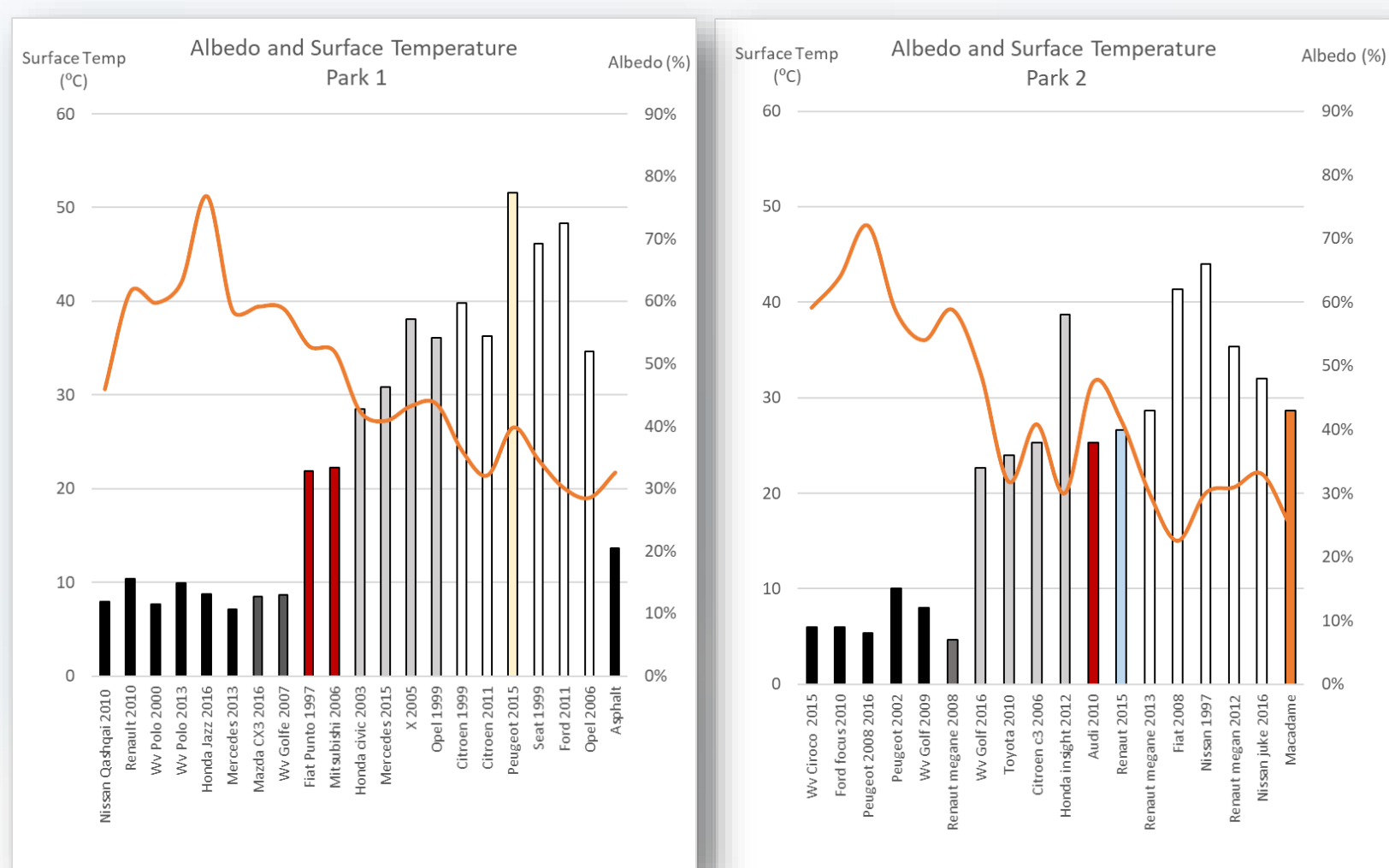
7 Jun measuring time: 12:00h-15:00h

Park 1
Air Temp: 33.2°C
RH: 20.4%
u max: 2.6 m/s
u: 1.5m/s

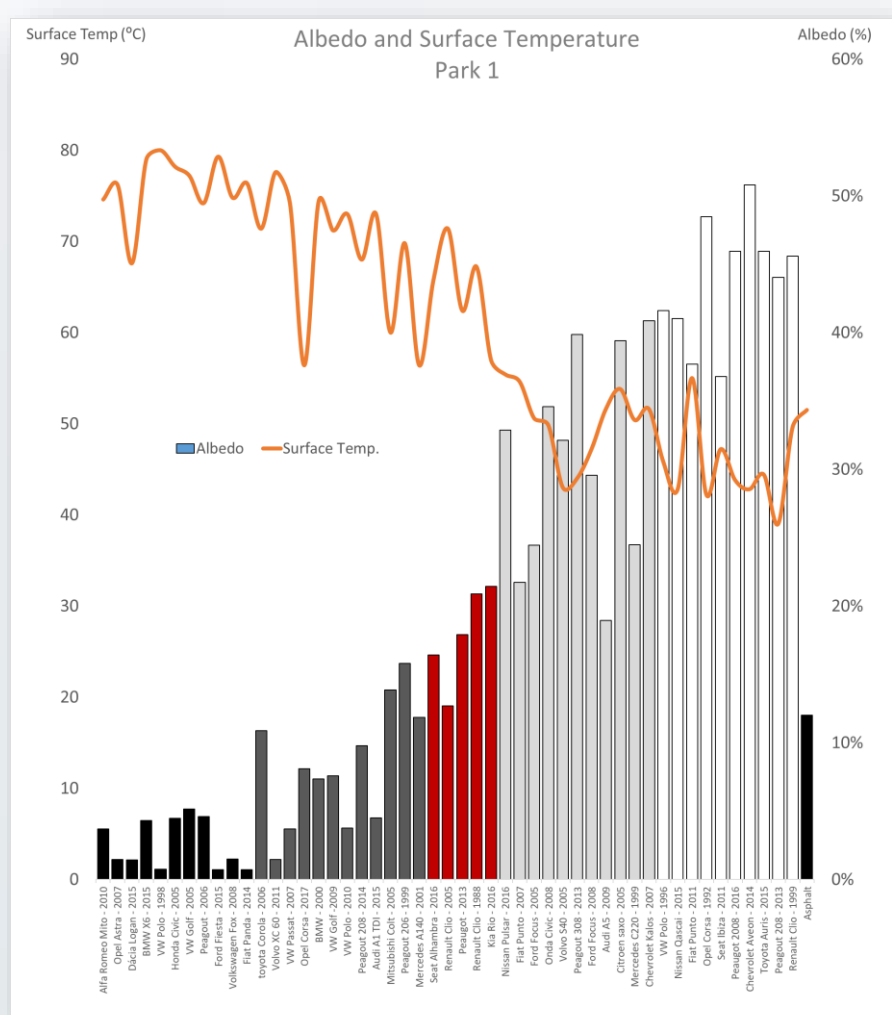


III - First results: Surface Temperatures and Albedo

16th November 2016



7th June 2017



IV Conclusions



the color of the bars are approximate to true vehicle colors

- Vehicles surfaces are very different from common walls and pavements and other elements like urban furniture and vegetation;
- About 7 % of the 2D mobile surfaces in Lisbon have never been taken into account in urban climate studies;
- If we consider 3D environment (the total exposed surfaces that interact with the urban canopy layers), these figures are much higher;
- At noon, white painted cars are highly reflective, with albedo between 40% and 50% and light grey between 20 and 40%;
- Surface temperatures are generally <30°C in Autumn and <40°C in the beginning of summer
- Black painted and dark grey cars have very absorptive surfaces (albedo <10%)
- Surface temperatures are generally > to 30/40°C in Autumn and > 60°C in the beginning of summer.

Ila – Methods and Data

How many vehicles are in Lisbon?



Method 1)

High Resolution
Image Classification



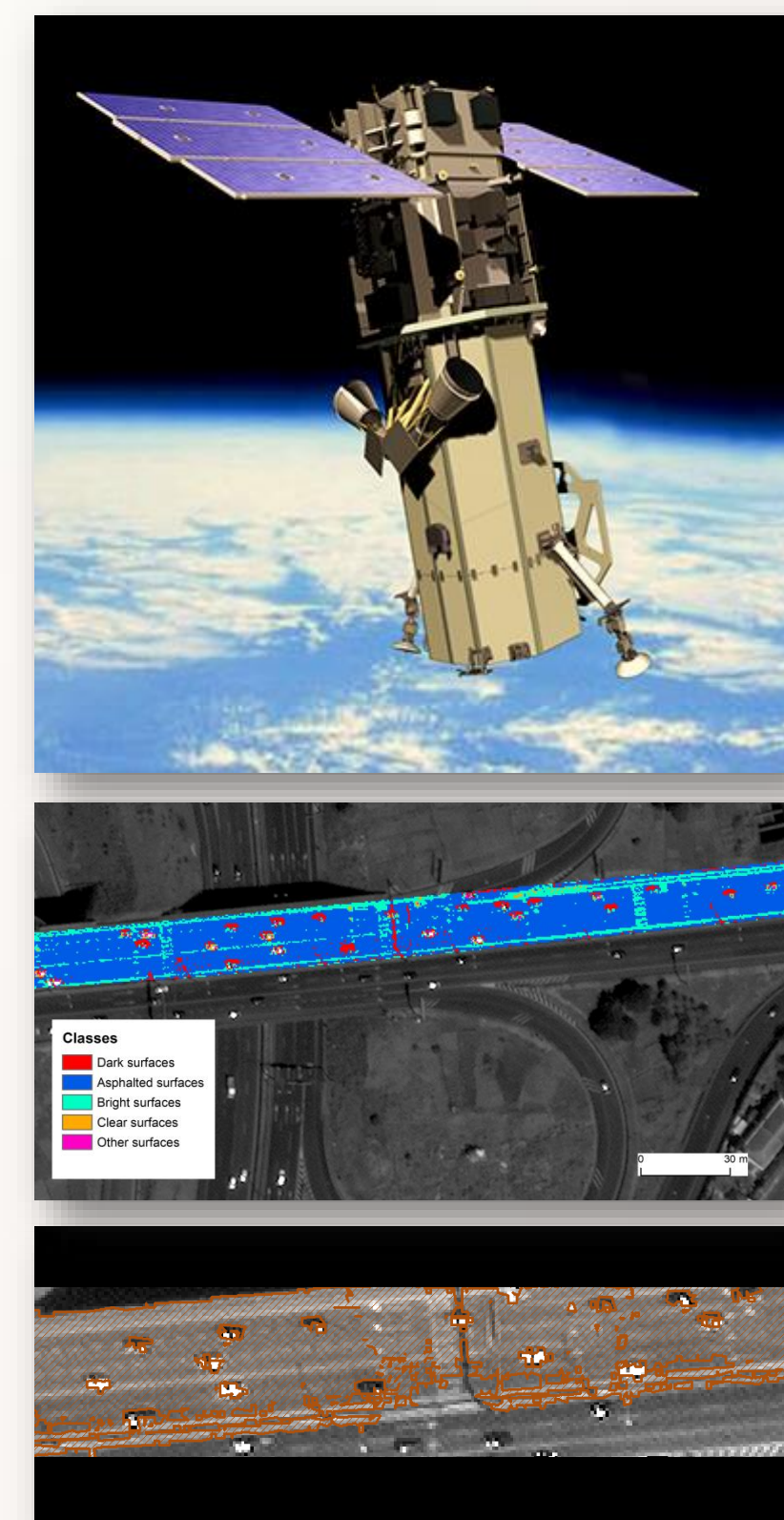
WorldView-2 (Digital Globe™) image set (0.5x0.5m resolution panchromatic band and 8 multispectral bands with 2x2m), from 30/04/2016 (10:30 am, LT)

Auxiliary information: green areas (NDVI); buildings and other important features as a mask.

a) After eliminating the vegetation and the buildings from the satellite image, this one was segmented using the “Segment Mean Shift Algorithm”.

b) The segments were used to define the training areas.

c) Supervised classification (Maximum Likelihood): definition of training areas, representing 5 classes (dark surfaces – black/grey vehicles, clear surfaces – Light color vehicles; asphalted surfaces, shiny surfaces, other surfaces).



Results:

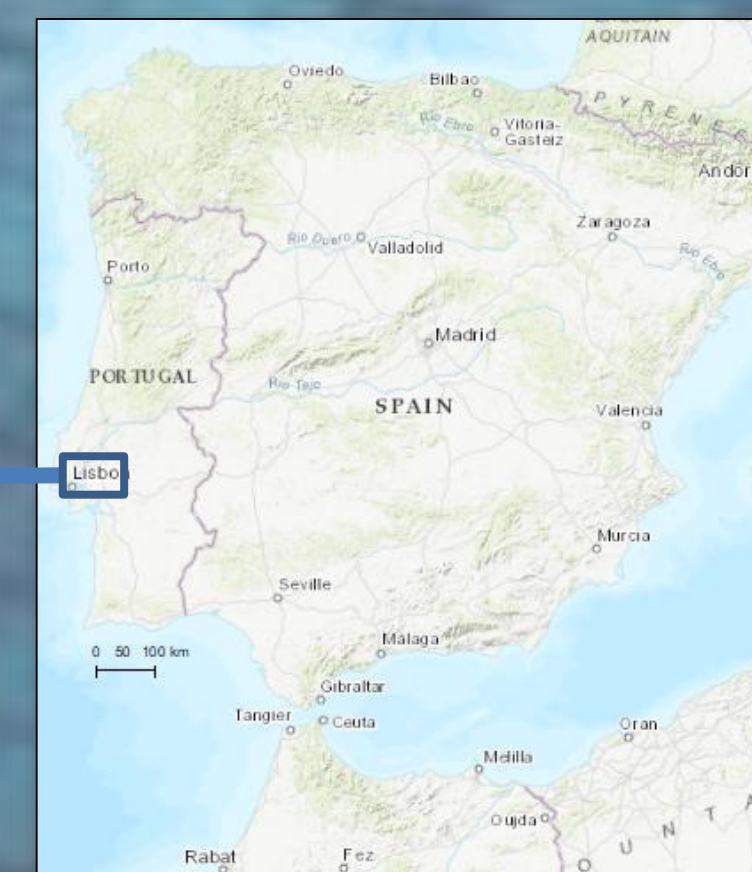
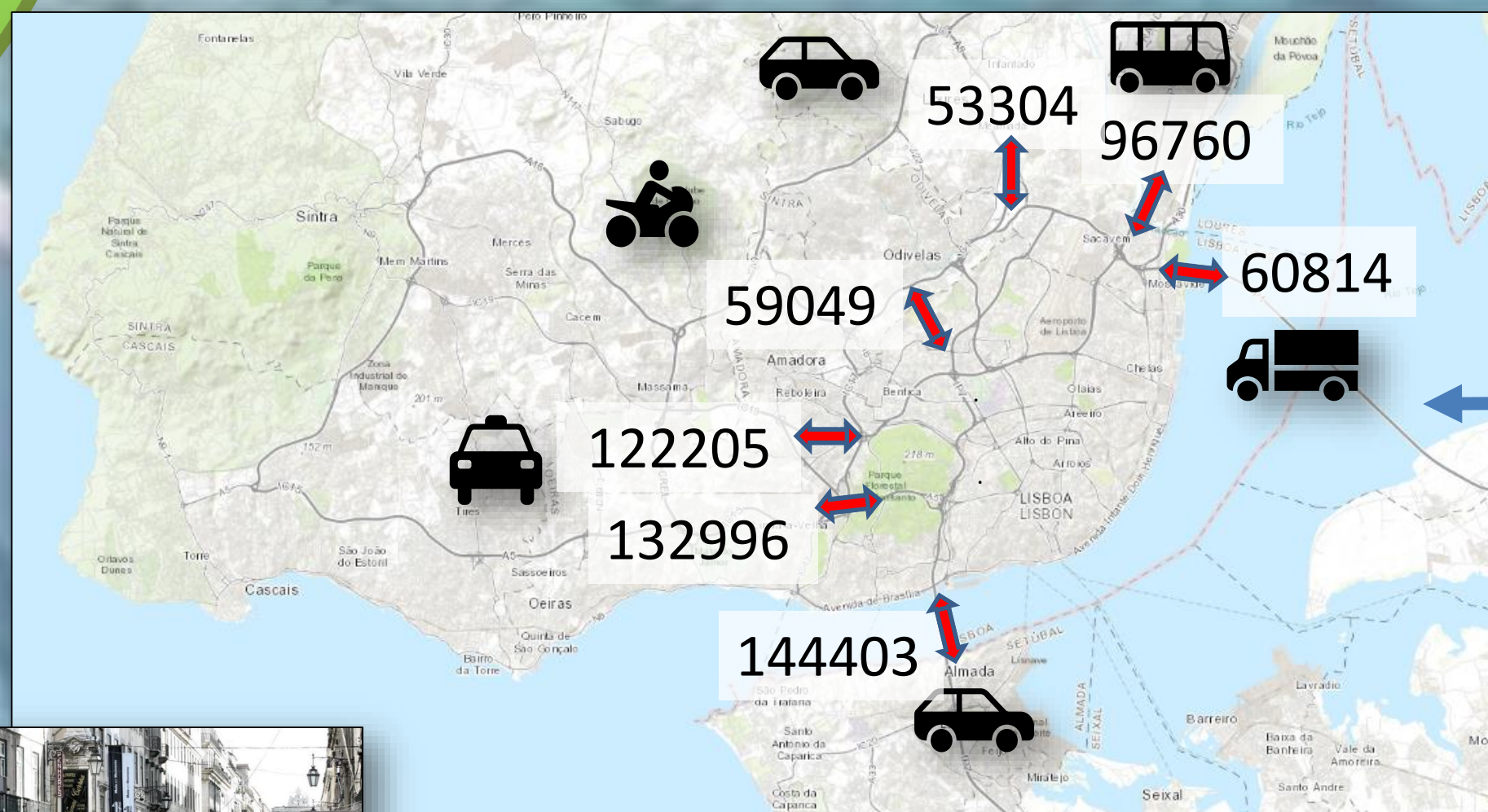
The vehicles occupy a total area of ≈6,6 km² (7.6% of the city area)



Method 2)

Car counting

All kind of vehicles entering and exiting Lisbon in the main highways (April to December, 2016)



Data source: *Instituto de Mobilidade e dos Transportes, IP*
Portuguese Institute of Mobility and Transportation, 2016.

Facts: Lisbon pop. – 547733

Lisbon MA pop. – 2821876

Data source: *Instituto Nacional de Estatística, Census 2011.*
Portuguese Institute of Statistics, 2018.

Total estimated entering the city: 370000 vehicles/day

Model: Portuguese ratios of vehicles / 1000 inhabitants (2015) were applied

Total estimated belonging to the city: 342500 vehicles/day

Total number of vehicles during the day = 712500

Model: Assuming an average 2D car dimension of about 8 m²

Results:

The vehicles occupy a total area of ≈5,7 km² (6.7% of the city area)

Color matters

(sample in the Saldanha neighborhood – city centre)

